

Mouhamed MBAYE

Natural Language Processing Engineer

Montpellier, France (Open to relocation)

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PROFILE

Specialized in Natural Language Processing (NLP) and an active member of the NLP team at [GalsenAI Lab](#), I am passionate about R&D in information retrieval and extraction. I lead research projects end to end and have hands-on experience with unstructured, multilingual, high-stakes business data.

TECHNICAL SKILLS

- . **Machine Learning & Deep Learning:** Python, PyTorch, Scikit-learn, Transformers, Fine-tuning
- . **Generative AI & LLMs:** RAG, Agentic AI, Prompt Engineering, vLLM, Ollama
- . **MLOps:** Docker, FastAPI, Flask, GCP, CI/CD, Git/GitHub, Streamlit
- . **Data & Vector DBs:** ChromaDB, PostgreSQL (pgvector), SQL, MySQL, MongoDB, Neo4j, SPARQL/RDF
- . **NLP R&D:** SpaCy, NLTK, Unitex-GramLab, Label-Studio, Hugging Face, Overleaf

PROFESSIONAL EXPERIENCE

NLP R&D Engineer

2026 – Present

GalsenAI Lab

- . Contributing to Wolof NLP research through the development and evaluation of LLMs, dataset design, and collaboration with senior researchers.

NLP Engineer

January – July 2025

IMGT (IGH, CNRS), Montpellier

- . Designed and deployed an information extraction solution for WHO scientific texts.

NLP Data Scientist

January – June 2024

Mohammed V University, Rabat

- . Automatic document modeling and classification.

PUBLICATIONS

August 2026

Accepted paper at NL4AI 2026: "Known but Unreachable: A Diagnostic Evaluation of Wolof Entity Linking against Wikidata".

Research paper on the diagnostic evaluation of Wolof entity linking against Wikidata, accepted for publication in the proceedings and for oral presentation at the **NL4AI 2026** workshop (co-located with AIXIA 2026 in Perugia, Italy).

[PDF](#)

May 2026

Preprint: "MudawanSn: A Gold-Standard Wolof–Arabic Parallel Corpus for Machine Translation".

Co-authored preprint (with Thierno Diop) introducing **MudawanSn**, a reference resource of **1,271** manually translated, sentence-aligned Wolof–Arabic pairs. Comparative evaluation of **NLLB-200**, **mT5** and **AfriNLLB**. The paper, dataset and code are publicly available.

[PDF](#)

EDUCATION

Master’s Degree (M2) in Natural Language Processing Université Marie & Louis Pasteur, Besançon, France	June 2025
Master’s Degree (M2) in Computational Linguistics Mohammed V University, Rabat, Morocco	June 2024

CERTIFICATIONS

Natural Language Processing with Probabilistic Models <i>DeepLearning.AI – 2024</i> . Probabilistic models, including N-grams, sequence labeling and autocomplete algorithms.
Natural Language Processing with Classification and Vector Spaces <i>DeepLearning.AI – 2024</i> . Core NLP techniques: sentiment analysis, vector spaces and word embeddings.
Retrieval Augmented Generation (RAG) <i>DeepLearning.AI – 2025</i> . Design, implementation and deployment of complete, reliable RAG systems tailored to specific application domains.
Text Mining for Marketing <i>O.P. Jindal Global University – 2025</i> . Foundations of text mining and best practices in text analysis for decision-making.
Elements of AI <i>University of Helsinki & MinnaLearn – 2024</i> . Fundamental AI concepts: machine learning, neural networks and societal implications.

LANGUAGES

French: Fluent
English: Advanced
Arabic: Fluent
Wolof: Native